FitHub AI

“Train smarter. Live better.”

Comprehensive Software Project Life Cycle Document

Project Name	FitHub AI
Application Type	AI-Powered Fitness Coaching Web Application
Primary Tech Stack	Python (FastAPI), Jinja2, SQLAlchemy, SQLite, Google Gemini API, Docker, Uvicorn

1. Brainstorming & Ideation Phase

Problem Statement

Standard workout applications rely heavily on static, pre-written templates that fail to account for real-time human variables such as equipment access, specific recovery states, body focus areas, or sudden daily time constraints.

Core Concept

FitHub AI provides an adaptive, real-time fitness coaching web application that automatically generates tailored workout routines and health advice built directly around individual user parameters.

Key Value Proposition

Users receive instantaneous, highly customized exercise routines and fitness coaching without needing expensive personal trainers.

Innovative Feature Concepts

- **Granular Input Personalization:** Accepts dynamic variables including recovery status, BMI goals, target weight, available equipment, and session length.
- **Generative AI Integration:** Leverages Google Gemini LLM (gemini-2.5-flash) for dynamic workout construction and daily fitness tips.
- **Modern Dashboard:** Interactive dark-mode user interface featuring visual progress rings, step counts, active minutes, and fitness scores.